

JACOB KLIMCZAK

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EXPERIENCE

DGP Lab, Machine Learning Researcher

May 2025 - Present

- Led a team of 5 students to build a fire-forecasting model, integrating insights from the Canadian Forest Service and Ontario Ministry of Natural Resources through weekly meetings, while keeping an active technical role.
- Extended Perceiver Encoder and SWIN Transformer-based model using **Python** and **PyTorch**, enabling the testing of 12 hypothetical fire scenarios, and digitally recreating Canada's biggest-ever wildfire.
- Designed modular pipeline infrastructure, reducing test load to 1/6th by creating shared model interfaces.

Freight Data Warehouse, Data Engineer

Jan 2023 - Present

- Led data initiatives as technical lead for a large Canadian transportation data repository, collaborating with 20+ partners including Transport Canada and the American Transportation Research Institute.
- Engineered 3 high-volume **Python** and **C#** data pipelines to anonymize and process **300+ GB/day** of sensitive data from diverse databases (**Postgres**, **MongoDB**, **MySQL**, **SQL Server**) on **VMWare/AWS** hybrid cloud.
- Scaled up emissions models for Transport Canada, optimizing pipelines to process 100x more data.
- Introduced **Python** GIS tooling using **genetic algorithms** to automate land use map creation for urban planners.

Intel/Altera, Software Engineer Intern

May 2024 - Sep 2025

- Implemented EDA features in a multi-million line **C++** codebase reducing customer ramp time by a month.
- Built Boost index data structures, cutting hierarchical filtering from minutes to ~20 ms on 10+ GB chip designs.
- Resolved **Qt**-based **memory corruption bugs** using **VTune** and **Valgrind**, improving platform stability by 30%.
- Migrated 2000+ legacy **Perl** tests to new testing framework expanding test coverage of new platform by 80%.

Temerity Analytics Inc., Full-Stack Developer

May 2023 - Feb 2024

- Expanded **Angular** and **Django**-based platform displaying locations and fleet performance for ~150 vehicles.
- Developed **Postgres** data solutions and sharded database, reducing speeding detection latency by 50% and enabling real-time detection.
- Implemented ~150hz **Flink** streaming logic for event-time vehicle tracking and fleet-wide metric aggregation.

Rotogran International Inc., Full-Stack Developer

Nov 2019 - Jan 2023

- Decomposed monolithic systems into containerized microservices with **Docker**, enabling **CI/CD** pipelines with **Github Actions** that reduced deployment cycle from monthly to weekly.
- Implemented **C#** pipelines digitizing paper records via OCR, cutting document retrieval from minutes to seconds.
- Developed **C#** and **Java** SDKs wrapping APIs, and using these in Windows, Android, and web (**React**) clients.

EDUCATION

University of Toronto, Computer Science

Sep 2021 - Jun 2026

- Dean's List scholar in 2022 & 2023, recipient of University of Toronto Scholar scholarship in 2021, recipient of Ruth Joyce Norton scholarship in 2022, recipient of Isabel Bader scholarship in 2023.
- Won **CLUE Symposium Best Paper in Data & Modeling** in 2025.
- Won **International CRA Outstanding Undergraduate Researcher** award honorable mention in 2025.
- Placed top 3 in the **Runway Ready** challenge with Dexterra in 2025 after creating machine learning models for Pearson International Airport in Toronto. Later funded by Dexterra to enhance the original prototype.
- Won **U of T IMI Big Data and AI Competition** in 2025 for developing a bank customer foundation model created with a **contrastive learning** approach; and applying it to **anomaly detection** models to find instances of money laundering. The model's performance led Scotiabank to test the model for internal use.